For full credit explain your reasoning, showing all relevant work.

Exercise 1. Let G be a group. Define a new group G' which has the same elements as G, but such that the multiplication in G' is given by

$$\underbrace{a \odot b}_{\text{multiplication in } G'} = \underbrace{b \cdot a}_{\text{multiplication in } G}$$

Show that the group G' is isomorphic to G.

Exercise 2. Let $\mathbb{R}$ be the group of real numbers with addition and $\mathbb{R}^*$ the group of non-zero real numbers with multiplication. Show that $\mathbb{R}$ is not isomorphic to $\mathbb{R}^*$.

Exercise 3. Show that the group $\mathbb{Z}_9$ is not isomorphic to the group $\mathbb{Z}_3 \times \mathbb{Z}_3$.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. Show that the group $\mathbb{Z}$ of integers is not isomorphic to the group $\mathbb{Q}$ of rational numbers.

Practice Exercise 2. Let $\mathbb{Q}$ be the group of rational numbers with addition and let $f: \mathbb{Q} \to \mathbb{Q}$ be a group homomorphism.

a) Show that for any $q \in \mathbb{Q}$ we have

$$f(q) = q \cdot f(1)$$

b) Show that if $\mathrm{Ker}(f) \neq \mathbb{Q}$ then f is an isomorphism.

Practice Exercise 3. Let G be a group. Consider a homomorphism $f: G \to G \times G$ given by $f(a) = (a, a)$. Show that the group G is abelian if and only if $\mathrm{Im}(f)$ is a normal subgroup of $G \times G$.

Practice Exercise 4. Let G be a finite abelian group which has no elements of order 2. Show that the function $f: G \to G$ given by $f(a) = a^2$ is an automorphism of G.